Lakshya JEE 2.0 2025

Mathematics

Indefinite Integration

DPP: 6

Q1 $\int \frac{2x-1}{(x-1)(x+2)(x-3)} dx$

(A) $\frac{1}{6}\log|x-1| - \frac{1}{3}\log|x+2| + \frac{1}{2}\log|x-3| + c$

(B) $\frac{-1}{6}\log|x-1| + \frac{1}{3}\log|x+2| + \frac{1}{2}\log|x-3| + c$

(C) $\frac{-1}{6}\log|x-1| - \frac{1}{3}\log|x+2| - \frac{1}{2}\log|x-3| + c$

(D) $\frac{-1}{6}\log|x-1| - \frac{1}{3}\log|x+2| + \frac{1}{2}\log|x-3| + c$

Q2 $\int \frac{x^4}{(x+2)(x^2+1)} dx$

(A) $\frac{x^2}{2} - 2x - \frac{16}{5}\log(x+2) - \frac{1}{10}\log(x^2+1) + \frac{2}{5}\tan^{-1}x + c$

(B) $\frac{x^2}{2} + 2x + \frac{16}{5}\log(x+2) - \frac{1}{10}\log(x^2+1) + \frac{2}{5}\tan^{-1}x + c$

(C) $\frac{x^2}{2} - 2x + \frac{16}{5}\log(x+2) - \frac{1}{10}\log(x^2+1) - \frac{2}{5}\tan^{-1}x + c$

(D) $\frac{x^2}{2} - 2x + \frac{16}{5}\log(x+2) - \frac{1}{10}\log(x^2+1) + \frac{2}{5}\tan^{-1}x + c$

Q3 $\int \frac{x^2+2}{(x+1)(x^2-1)} dx$

(A) $\frac{1}{4}\log|x+1| + \frac{3}{2(x+1)} + \frac{3}{4}\log|x-1| + c$

(B) $\frac{1}{4}\log|x+1| - \frac{3}{2(x+1)} + \frac{3}{4}\log|x-1| + c$

(C) $\frac{1}{4}\log|x+1| + \frac{3}{2(x+1)} - \frac{3}{4}\log|x-1| + c$

(D) $-\frac{1}{4}\log|x+1| + \frac{3}{2(x+1)} + \frac{3}{4}\log|x-1| + c$

Q4 $\int \frac{x^2+x-6}{(x-2)(x-1)} dx$ is equal to:

(A) $x + 2\log(x-1) + c$

(B) $2x + 2\log(x-1) + c$

(C) $x + 4\log(1-x) + c$

(D) $x + 4\log(x-1) + c$

Q5 $\int \sqrt{4-x^2}\, dx$

Q6 $\int \frac{(x+x^3)^{1/3}}{x^4} dx$ is equals to

(A) $\frac{3}{8}\left(\frac{1}{x^2}-1\right)^{4/3} + C$

(B) $-\frac{3}{8}\left[\left(1+\frac{1}{x^2}\right)^{4/3}\right] + C$

(C) $\frac{1}{8}\left(1+\frac{1}{x^2}\right)^{4/3} + C$

(D) $\frac{1}{8}\left(\frac{1}{x^2}-1\right)^{4/3} + C$

Q7 $\int \frac{dx}{x^5+x}$

(A) $\frac{1}{4}\log\left(1+\frac{1}{x^4}\right) + c$

(B) $\frac{1}{2}\log\left(1+\frac{1}{x^4}\right) + c$

(C) $-\frac{1}{2}\log\left(1+\frac{1}{x^4}\right) - c$

(D) $-\frac{1}{4}\log\left(1+\frac{1}{x^4}\right) + c$

Q8 Let $\int \frac{dx}{x^{2008}+x} = \frac{1}{p}\ln\left(\frac{x^q}{1+x^r}\right) + C$, where $p, q, r \in$ and need not be distinct then the value of $(p+q+r)$ equals:

(A) 6024
(B) 6022
(C) 6021
(D) 6020

Q9 $\int \frac{x^7}{\left(1-x^2\right)^5} dx$

(A) $\frac{1}{4} \cdot \frac{1}{\left(x^{-2}-1\right)^4}+c$

(B) $\frac{1}{6} \cdot \frac{1}{\left(x^{-2}-1\right)^4}+c$

(C) $\frac{1}{8} \cdot \frac{1}{\left(x^{-2}-1\right)^4}+c$

(D) $\frac{1}{12} \cdot \frac{1}{\left(x^{-2}-1\right)^4}+c$

Q10 $\int \frac{dx}{x^2\left(1+x^5\right)^{4/5}}=-\left(1+\frac{1}{x^m}\right)^n+c$. Find mn

Q11 The value of $\int \frac{dx}{x^2\left(x^4+1\right)^{3/4}}$

(A) $\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$

(B) $\left(x^4+1\right)^{1/4}+c$

(C) $-\left(x^4+1\right)^{1/4}+c$

(D) $-\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$

Q12 $\int \frac{2x^{12}+5x^9}{\left(x^5+x^3+1\right)^3} dx$

(A) $\frac{-x^5}{\left(x^5+x^3+1\right)^2}+c$

(B) $\frac{x^{10}}{2\left(x^5+x^3+1\right)^2}+c$

(C) $\frac{x^5}{2\left(x^5+x^3+1\right)^2}+c$

(D) $\frac{-x^{10}}{2\left(x^5+x^3+1\right)^2}+c$

Q13 $\int x^{-7}\left(1+x^4\right)^{-1/2} dx$ is equal to.

(A) $\frac{\sqrt{x^4+1}\left(2x^4-1\right)}{6x^6}+c$

(B) $\frac{\sqrt{x^4+1}}{6x^6}+c$

(C) $\frac{2x^4-1}{6x^6}+c$

(D) None of these

Q14 $\int \frac{dx}{(x-2)^{7/8}(x+3)^{9/8}}=$

(A) $\frac{8}{5}\left(\frac{x-2}{x+3}\right)^{1/8}+c$

(B) $\frac{5}{8}\left(\frac{x-2}{x+3}\right)^{1/8}+c$

(C) $\frac{5}{8}\left(\frac{x+3}{x-2}\right)^{1/8}+c$

(D) $\frac{8}{5}\left(\frac{x+3}{x-2}\right)^{1/8}+c$

Q15 $\int \frac{\sin^3 x dx}{\left(\cos^4 x+3\cos^2 x+1\right)\cdot\tan^{-1}(\sec x+\cos x)}=$

(A) $\tan^{-1}(\sec x+\cos x)+c$

(B) $\log\left(\tan^{-1}(\sec x+\cos x)\right)+c$

(C) $\frac{1}{\sec x+\cos x}+c$

(D) None of these

Q16 $\int \frac{\left(x^2-1\right) dx}{\left(x^4+3x^2+1\right)\tan^{-1}\left(\frac{x^2+1}{x}\right)}=\ln|f(x)|+C$

Then $f(x)$ is:

(A) $\ln\left(x+\frac{1}{x}\right)$

(B) $\tan^{-1}\left(x+\frac{1}{x}\right)$

(C) $\cot^{-1}\left(x+\frac{1}{x}\right)$

(D) $\ln\left(\tan^{-1}\left(x+\frac{1}{x}\right)\right)$

Q17 The evaluation of $\int \frac{px^{p+2q-1}-qx^{q-1}}{x^{2p+2q}+2x^{p+q}+1} dx$ is:

(A) $-\frac{x^p}{x^{p+q}+1}+C$

(B) $\frac{x^q}{x^{p+q}+1}+C$

(C) $-\frac{x^q}{x^{p+q}+1}+C$

(D) $\frac{x^p}{x^{p+q}+1}+C$

Q18 $\int \frac{1-x^2}{1+x^2} \cdot \frac{dx}{\sqrt{1+x^2+x^4}}$

(A) $\operatorname{cosec}^{-1}\left(x+\frac{1}{x}\right)+c$

(B) $-\operatorname{cosec}^{-1}\left(x+\frac{1}{x}\right)+c$

(C) $\cos^{-1}\left(x+\frac{1}{x}\right)+c$

(D) $-\cos^{-1}\left(x+\frac{1}{x}\right)+c$

Q19 If $f(x)=\int \frac{5x^8+7x^6}{\left(x^2+1+2x^7\right)^2} dx, \quad (x \geq 0), f(0)$
$=0$
and $f(1)=\frac{1}{k}$, then the value of k is:

Q20 $\int \frac{2x^3-1}{x^4+x} dx$

(A) $-\log\left(x^3+1\right)-\log x+c$

(B) $\log\left(x^3+1\right)+\log x+c$

(C) $\log\left(x^3+1\right)-\log x+c$

(D) $-\log\left(x^3+1\right)+\log x+c$

Q21 $\int \frac{dx}{(x+4)^{8/7}(x-3)^{6/7}}$

(A) $\left(\frac{x-3}{x+4}\right)^{2/7}+c$

(B) $-\left(\frac{x-3}{x+4}\right)^{1/7}+c$

(C) $\left(\frac{x-3}{x+4}\right)^{1/7}+c$

(D) $-\left(\frac{x-3}{x+4}\right)^{2/7}+c$

Q22 If

$\int \frac{\cos x d x}{\sin ^3 x\left(1+\sin ^6 x\right)^{2/3}}=f(x)\left(1+\sin ^6 x\right)^{1/\lambda}+c$,where

c is a constant of integration, then $\lambda \cdot f\left(\frac{\pi}{3}\right)=$

(A) $\frac{9}{8}$

(B) 2

(C) -2

(D) $-\frac{9}{8}$

Q23 $\int \frac{x^2-1}{x \sqrt{x^4+3 x^2+1}} d x$ equals

(A) $\log \left|\frac{x^2+1+\sqrt{x^4+3 x^2+1}}{x}\right|+C$

(B) $\log \left|\frac{x+1+\sqrt{x^2+3 x+1}}{x}\right|+C$

(C) $\log \left|\frac{x^2+1+\sqrt{x^2+1}}{x^2}\right|+C$

(D) $\tan ^{-1} x+C$

Answer Key

Q1	(D)
Q2	(D)
Q3	(A)
Q4	(D)
Q5	$\frac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\frac{x}{2}+c$
Q6	(B)
Q7	(D)
Q8	(C)
Q9	(C)
Q10	1
Q11	(D)
Q12	(B)
Q13	(A)
Q14	(A)
Q15	(B)
Q16	(B)
Q17	(C)
Q18	(A)
Q19	4
Q20	(C)
Q21	(C)
Q22	(C)
Q23	(A)